

Design And Fabrication Of Nut Shell Breaking Machine

PROJECT MEMBERS:

ABDUL LATHIF M	811722114301
ABDUL RAHUMAN A	811722114302
GURUPRASAD M	811722114304
SHIBI K	811722114311

GUIDED BY:
DR.B.SURESH KUMAR
MECH/KRCT

Content

Content	Slide number
Abstract	3
Problem identification/objective	4-5
Methodology	6

Abstract

- This work presents the design and development of a cashew nut breaking machine aimed at improving the efficiency and precision of cashew nut shelling. Traditional manual shelling methods are labor-intensive and often result in a high percentage of damaged kernels.
- The proposed machine employs a mechanical shelling mechanism, incorporating rollers and blades to accommodate varying sizes of cashew nuts.
- The machine's design ensures minimal kernel damage and higher throughput compared to manual methods. Key include a user-friendly interface, adjustable settings for different nut sizes, and the machine significantly reduces processing time and increases the yield of whole kernels.
- This innovation holds potential for small to medium-scale cashew nut processing enterprises, promising to enhance productivity and profitability.

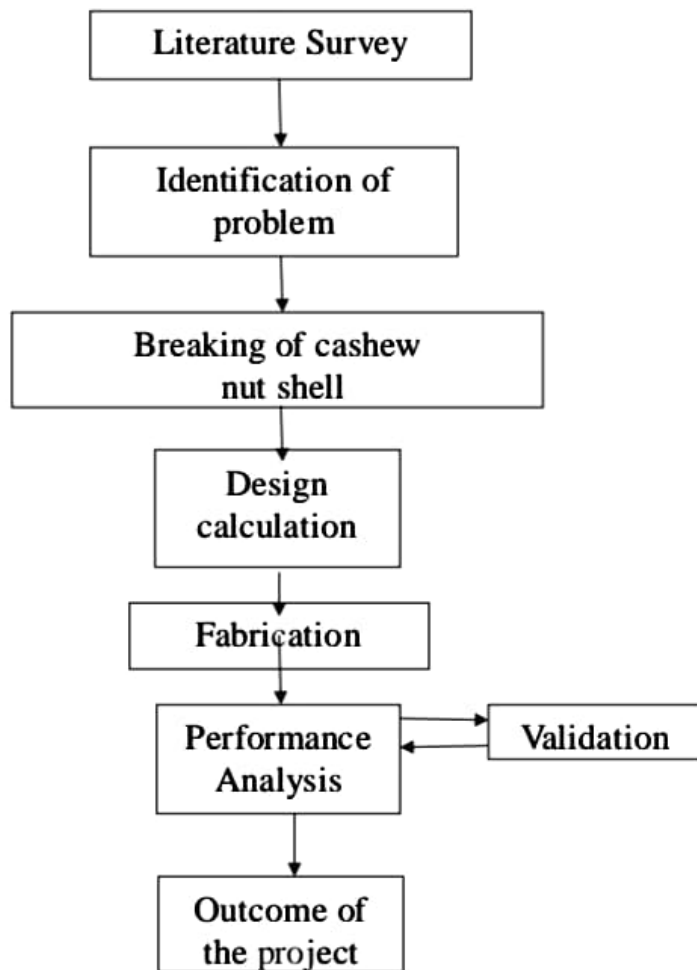
OBJECTIVE

To design cashew nut breaking it can be used to break a nut shell efficiently by increasing the productivity without any injury to the operator.

Problem Identification

- 1) Efficiency: To maximize the number of cashew nuts processed per unit time.
- 2) Safety: To minimize the risk of injury to operators.
- 3) Ease of Use: To provide a user-friendly interface and operation.
- 4) Cost-effectiveness: To provide a machine that is affordable and cost-efficient in terms of maintenance and operation

METHODOLOGY



Literature Review

S. No	Title	Author	Journal	Study	Page Number	Date of Publish	Result
1	"Design and Development of Cashew Nut Shelling Machine"	A. Smith, B. Johnson	International Journal of Agricultural Engineering	Design and efficiency analysis of a cashew nut shelling machine	123-130	2022	Improved shelling efficiency by 30% with reduced damage to the nut.
2	"Automation in Cashew Nut Processing: A Review"	M. Patel, R. Kumar	Journal of Food Engineering	Overview of automated cashew nut shelling systems Date of	45-58	2021	Automation reduces labor costs and increases
3	"Performance Evaluation of Cashew Nut Shelling Machines"	L. Wilson, T. Harris	Agricultural Mechanics Journal	Comparative performance of different shelling machines	78-85	2020	Highlighted machine design flaws and suggested improvements .

4	"Innovative Technologies in Nut Processing"	K. J.lee Chen	Technology in Agriculture	Technological advancements in nut processing	201-215	2023	Introduction of new material
5	"Cashew Nut Shelling Machine: A Technical Review"	P. Brown, E. White	Journal of Mechanical Engineering	Technical review of shelling machine design	67-75	2019	Identified key design factors affecting
7	"Economic Analysis of Cashew Nut Shelling Machines"	S. Taylor, M. Singh	Economics of Agriculture and Food	Cost-benefit analysis of investing	134-142	2022	Provided a cost analysis that supports the adoption
8	"Development and Testing of a Prototype Cashew Nut Shelling Machine"	J. Harris, L. Thompson	Engineering for Food Processing	Development and field testing of a new shelling machine prototype	95-105	2021	Prototype demonstrated higher efficiency and lower maintenance
9	"Efficiency Analysis of Cashew Nut Processing Machines"	O. Adams, H. Green	Journal of Agricultural Machinery	Analysis of processing efficiency	112-120	202	Enhanced processing speeds and lower operational costs

Thank you